

SafetyEye – AI Powered Workplace Occupancy & Safety Monitor

Milestone 2 Report

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Program: Infosys Virtual Internship

Project: SafetyEye – AI Powered Workplace Occupancy & Safety Monitor

Framework: YOLOv8 (Ultralytics)

Hardware Used: NVIDIA GTX 1650 Ti GPU (4GB)

Software: Python, PyTorch, Ultralytics YOLOv8, OpenCV, VS Code

Technical Stack

- **Framework:** YOLOv8 (Ultralytics)
 - **Programming Language:** Python
 - **Libraries:** PyTorch, OpenCV
 - **Development Environment:** VS Code
 - **Hardware:** NVIDIA GTX 1650 Ti GPU (4GB)
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Milestone 2 Objectives

The primary objectives of Milestone 2 were:

- Perform full model training using the prepared dataset
 - Evaluate model performance using standard detection metrics
 - Improve model performance through experimentation and hyperparameter tuning
 - Identify the best-performing model configuration
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Week 3 – Model Training and Evaluation

Objective

To train the YOLOv8 model and evaluate its performance using key object detection metrics.

Model Training

The **YOLOv8 Nano (YOLOv8n)** model was trained using GPU acceleration.

Training Configuration

Parameter Value

Model	YOLOv8n
Epochs	50
Image Size	640
Hardware	NVIDIA GTX 1650 Ti

Training Results

Metric	Value
Precision	0.857
Recall	0.683
mAP@0.5	0.763
mAP@0.5:0.95	0.506

These results showed a significant improvement over the initial training from Milestone 1.

Metrics Explanation

- Precision (0.857):** Indicates the accuracy of predicted detections
 - Recall (0.683):** Measures how many actual objects were detected
 - mAP@0.5 (0.763):** Overall detection performance at IoU = 0.5
 - mAP@0.5:0.95 (0.506):** Bounding box accuracy across multiple IoU thresholds
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Model Validation

- Model was validated on the validation dataset
 - Performance remained stable across multiple object classes
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Test Predictions

The trained model successfully detected:

- Person
 - Helmet
 - Safety Vest
 - Machinery
 - Safety Cone
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Week 4 – Model Optimization

Objective

To improve model performance using hyperparameter tuning and experimentation.

Experiments and Results

Experiment 1 – Increased Epochs

Parameter Value

Model YOLOv8n

Epochs 100

Image Size 640

Results

Metric	Value
Precision	0.873
Recall	0.723
mAP@0.5	0.795

Metric	Value
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mAP@0.5:0.95	0.542
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Observation: Increasing epochs improved learning and overall performance.

Experiment 2 – Increased Image Resolution

Parameter	Value
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Model	YOLOv8n
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Epochs	50
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Image Size	800
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Results

Metric	Value
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Precision	0.845
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Recall	0.685
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mAP@0.5	0.766
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mAP@0.5:0.95	0.484
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Observation: Higher resolution increased computational cost but gave minimal accuracy improvement.

Experiment 3 – Larger Model

Parameter	Value
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Model	YOLOv8s
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Epochs	50
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Image Size	640
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Results

Metric	Value
Precision	0.885
Recall	0.730
mAP@0.5	0.802
mAP@0.5:0.95	0.536

Observation: Larger model achieved the best performance.

Data Augmentation Experiment

Techniques Used

- Mosaic Augmentation
- Horizontal Flip
- Brightness Adjustment
- Saturation Variation

Results

Metric	Value
Precision	0.874
Recall	0.727
mAP@0.5	0.803
mAP@0.5:0.95	0.536

Observation: Improved generalization with slight performance gain.

Final Model Selection

Parameter Value

Model YOLOv8s

Epochs 50

Image Size 640

Precision 0.885

Recall 0.730

mAP@0.5 0.802

Error Analysis

Observed limitations:

- Small objects (e.g., gloves) sometimes missed
 - Helmet detection failed under occlusion
 - Overlapping bounding boxes in crowded scenes
 - Reduced accuracy in low-light conditions
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Possible Improvements

- Increase dataset size
 - Add more small object samples
 - Use higher image resolution strategically
 - Train with larger models (e.g., YOLOv8m)
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Experiment Comparison Table

Experiment	Model	Epochs	Img Size	Precision	Recall	mAP@0.5	mAP@0.5:0.95
Baseline (Week 3)	YOLOv8n	50	640	0.857	0.683	0.763	0.506
Experiment 1	YOLOv8n	100	640	0.873	0.723	0.795	0.542
Experiment 2	YOLOv8n	50	800	0.845	0.685	0.766	0.484
Experiment 3	YOLOv8s	50	640	0.885	0.730	0.802	0.536
Augmentation	YOLOv8s	50	640	0.874	0.727	0.803	0.536

Milestone 2 Outcomes

- Completed full YOLOv8 model training
- Evaluated model using precision, recall, and mAP
- Conducted multiple optimization experiments
- Identified best-performing model configuration

Conclusion

Milestone 2 focused on model training, evaluation, and optimization. The YOLOv8 model was successfully trained and analyzed using standard detection metrics.

Through systematic experimentation—including tuning epochs, image resolution, model size, and augmentation—the **YOLOv8s model (50 epochs, 640 image size)** was identified as the optimal configuration.

This model provides strong detection accuracy and will be used in the next phase for **real-time safety monitoring and violation detection**.